

Linear Algebra—Winter/Spring 2026

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Problem Set 7

(Updated: Friday, March 13, 2026)

Due: 8 PM, Tuesday March 17, 2026

Instructions

- a. *Timeline.* In general, expect the first “half” of a problem set to be released by Tuesday nights (or whenever the first lecture of the week takes place). Expect the full problem set to be released by Friday nights (or whenever the second lecture of the week takes place). The full assignment will, *typically*, be due on Tuesday nights, 8 PM.
- b. *Punctuality.* Aim to get the assignment done three days prior to the deadline. We will not be able to entertain any requests for extensions. If you seek clarification from us and we do not respond on time, do the following: (a) assume whatever seems most reasonable, and solve the question under that assumption and (b) ensure you have evidence for making the clarification request and report the incident to one of the PhD TAs.
- c. *Resources.* Feel free to use any resource you like (including generative AI tools) to get help but please make sure you understand what you finally write and submit. The TAs may ask you to explain the reasoning behind your response if something appears suspicious.
- d. *Graded vs Practice Questions.* Problems numbered using Arabic numerals (i.e. 1, 2, 3, ...) constitute assignment problems that will be graded. Problems numbered using Roman numerals (i.e. i, ii, iii, ...) are practice problems that will not be graded.

The text here is Linear Algebra Done Right, by Sheldon Jay Axler (3rd edition); we are on Chapter 5 (but basic material has been taken from Chapter 3).

1. Let T be a linear operator.
 - a. Show that T has a 1-dimensional invariant subspace iff T has an eigenvalue.
 - b. Show that $\text{range}(T)$ is an invariant subspace of T .
 - c. Show that $\text{nullspace}(T)$ is an invariant subspace of T . Given that T has an eigenvalue λ , what can you say about $\text{nullspace}(T - \lambda\mathbb{I})$?
2. Let V be a vector space and let $S, T \in \mathcal{L}(V)$ be linear operators such that $ST = TS$. Prove that the null space of S and the range of S are invariant under T .

3. Let $P(\mathbb{R})$ denote the vector space of all real polynomials. Define the linear transformation $T : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ by $T(p) = p'$, where p' denotes the derivative of p . Find all eigenvalues and the corresponding eigenvectors of T .
4. Consider the vector space $V := \mathbb{R}^2$. Consider the vector $v(\theta) \in V$ defined as $v(\theta) := r \cdot (\cos \theta, \sin \theta)$.
 - a. With $(1, 0)$ along the x -axis and $(0, 1)$ along the y -axis, schematically depict the vector $v(\theta)$ in a figure.
 - b. Suppose T is a transformation that rotates all vectors in $\mathbb{R}^2$ (counter clock-wise) by an angle ϕ . Write down an expression for $Tv(\theta)$.
 - c. Show that for $\phi = \pi/2$, a vector $v = (a, b)$ gets mapped to $Tv = (-b, a)$.
 - d. Based on this geometric intuition, give an informal argument that shows that a rotation in 2 dimensions, cannot have a real eigenvalue. Assume $\phi = \pi/2$.
 - e. Prove this fact algebraically, using the operator $T - \lambda \mathbb{I}$. Assume $\phi = \pi/2$.
5. (This asks you to essentially prove Proposition 5.26 in the text)
 Suppose $T \in L(V)$ and $v_1, \dots, v_n$ is a basis of V . Show that the following are equivalent.
 - a. The matrix of T with respect to $v_1, \dots, v_n$ is upper triangular.
 - b. $Tv_j \in \text{span}(v_1, \dots, v_j)$ for each $j \in \{1, \dots, n\}$.
 - c. $\text{span}(v_1, \dots, v_j)$ is invariant under T for each $j \in \{1, \dots, n\}$.
- i. Let V be a vector space and let $U_1, \dots, U_m$ be subspaces of V . Suppose that

$$U_1 + U_2 + \dots + U_m$$

is a direct sum. Prove that for every subset $S \subseteq \{1, 2, \dots, m\}$, the sum

$$\sum_{i \in S} U_i$$

is also a direct sum.

- ii. Let V be a vector space and let $U_1, \dots, U_m$ be subspaces of V . Suppose that

$$U_1 + U_2 + \dots + U_m$$

is a direct sum. Prove that for every pair $i \neq j$, $U_i \cap U_j = \{0\}$.

Is the converse true? That is, if $U_i \cap U_j = \{0\}$ for every $i \neq j$, does it follow that

$$U_1 + U_2 + \dots + U_m$$

is a direct sum? Prove the statement or give a counterexample.

(Hint: Does something interesting happen as you go from $m = 2$ to $m = 3$).

- iii. Suppose $U_1, \dots, U_m$ are subspaces of some vector space U . Show $U_1 + \dots + U_m$ is a direct sum iff the only way of writing 0 as a sum $\sum_i u_i$ (where $u_i \in U_i$) is by taking each $u_j = 0$.

(It is a proposition in the text somewhere; but try thinking about it yourself; the idea is very similar to the proof that 1:1 means $\dim(\text{kernel})$ is zero (for linear operators)).

- iv. (This is based on Example 1.43 in the text)

$$U_1 := \{(x, y, 0) : x, y \in F\}, U_2 := \{(0, 0, z) : z \in F\}, U_3 := \{(0, y, y) : y \in F\}.$$

See if you can write $(0, 0, 0)$ in two different ways as a sum $u_1 + u_2 + u_3$ where $u_1, u_2, u_3 \in U$. Also check for which i, j , it holds that $U_i \cap U_j = \{0\}$.

- v. Suppose $T \in \mathcal{L}(V)$ is a linear operator with λ as an eigenvalue. Let $U := \text{range}(T - \lambda I)$. Show that U is invariant under T .

6. [Based on Example 5.15] Suppose T is a linear operator on F^2 , defined by $T(x, y) = (0, x)$. Let $U := \{(0, y) : y \in F\}$.

- Show that all vectors $u \in U$ are eigenvectors of T with eigenvalue 0.
- Show that the only eigenvalue T can have is 0.
- Suppose v is an eigenvector of T . Show that $v \in U$.
- Using your answers, show that there does not exist a subspace W that is (i) invariant under T and (ii) is such that $F^2 = U \oplus W$.

7. Suppose $T \in \mathcal{L}(V)$ is a linear operator acting on the vector space V over the field of complex numbers. Prove the following about T .

- Let $v \in V$ be any non-zero vector. Show that there exist numbers $a_0, a_1, \dots, a_n$, not all zeros, such that $0 = \sum_{i \in \{0, \dots, n\}} a_i T^i v$, where n is the dimension of V .
- Show that $a_0, a_1, \dots, a_n$ satisfying the equation above, are such that $a_1, \dots, a_n$ cannot all be 0.
- Show that if $(T - \lambda_1 I) \dots (T - \lambda_m I)v = 0$ for $v \neq 0$, then $(T - \lambda_j I)$ is not injective, for at least one j .